

Image Enhancement?

☞ **Enhancement :** is to process an image so that the result is more suitable than the original image for a *specific* application.

Enhancing an image provides better contrast and a more detailed image as compare to non enhanced image. It is used to enhance medical images, images captured in remote sensing, images from satellite e.t.c

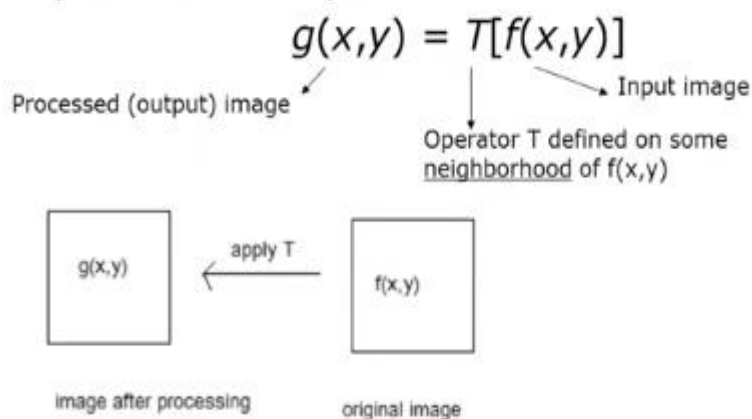
Enhancement techniques fall into 2 types:

- ▣ **Spatial domain:** direct manipulation of pixels in the image plane
- ▣ **Frequency domain:** modifying frequency spectrum of the image.

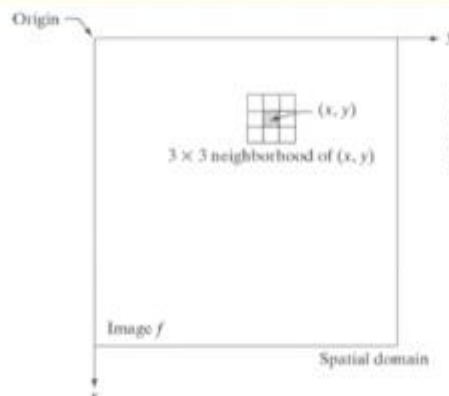
Enhancement in Spatial Domain

- ▣ **Spatial domain:** aggregate of pixels composing an image

Spatial domain processes:



Defining a Neighborhood (T)



Operator T is applied at each location (x, y) to produce the output g at that location.

types of neighborhood:

1. **intensity transformation**: neighborhood of size 1×1
2. **spatial filter** (or mask, kernel, template or window): neighborhood of larger size, like in the above example.

Intensity Transformations Functions

- Operation deals with pixel intensity values individually
- The transformed output pixel value does not depend on any of the neighboring pixel value of the input image
- The intensity values are altered using particular transformation techniques
- Here, T is called intensity transformation function or (mapping, gray level function)

$$\underbrace{g(x, y)}_s = T[\underbrace{f(x, y)}_r]$$

$$s = T(r)$$

where r is the pixels of the input image and s is the pixels of the output image

Spatial Filtering

- ▣ Mask is a small matrix useful for blurring, sharpening, edge-detection and more.
- ▣ New image is generated by applying the mask matrix on the input image.
- ▣ The output pixel values thus depend on the neighboring input pixel values.
- ▣ The mask may be of any dimension 3X3 4X4

Intensity transformations functions

Intensity transformation functions fall into 2 approaches:

1) Basic intensity transformations

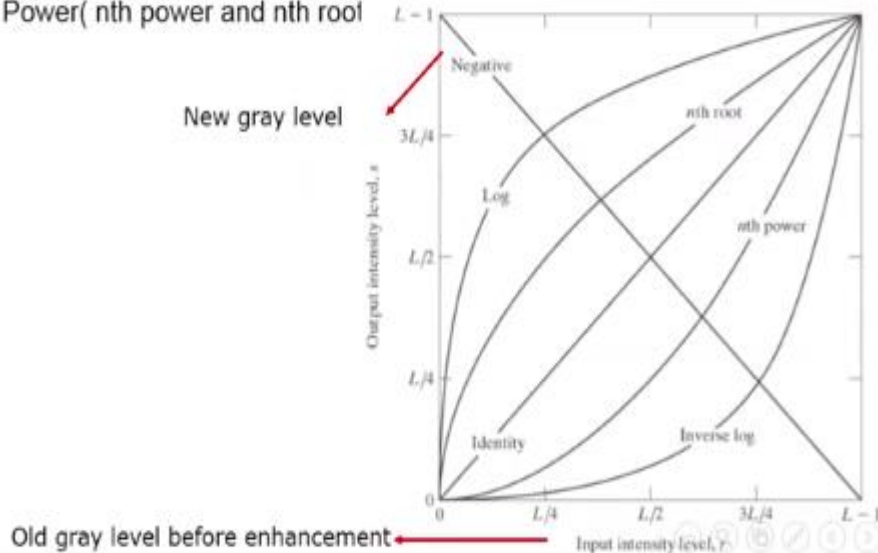
- a) Linear (negative and identity).
- b) logarithmic (Log and Inverse Log) .
- c) Power(nth power and nth root).

2) piecewise Linear transformation functions.

- a) Contrast stretching, thresholding
- b) Gray-level slicing
- c) Bit-plane slicing

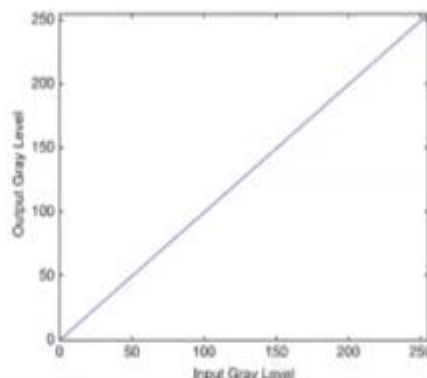
Basic Intensity (gray level) Transformations

- a) Linear (negative and identity).
- b) logarithmic (Log and Inverse Log) .
- c) Power(nth power and nth root)



Identity Transformation Function

- In **Identity transformation**, each value of the input image is directly mapped to each other value of output image.
- Thus, the output intensities are identical to the input intensities. It has been shown below:



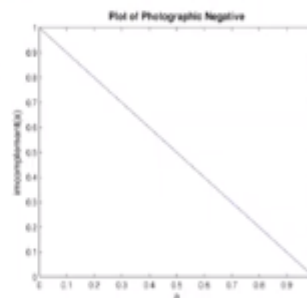
Negative Transformation Function

- ▣ **Negative images** are useful for enhancing white or grey detail embedded in dark regions of an image.
- ▣ The negative of an image with gray levels in the range $[0, L-1]$ is obtained by using the expression

$$s = L - 1 - r$$

$(L-1)$ = Maximum pixel value where L is the number of gray levels in the input image

r = input Pixel value of an original image



Negative Transformation Function

The negative of an image with intensity levels in the range $[0, L-1]$ is obtained by using the negative transformation :

$$s = L - 1 - r$$

Image (r)



Image (s) after applying T (negative)



Advantages of negative :

- ✓ Produces an equivalent of a photographic negative.
- ✓ Enhances white or gray detail embedded in dark regions especially when the black areas are dominant in size.

Negative Transformation Function

The negative of an image with intensity levels in the range $[0, L-1]$ is obtained by using the negative transformation :

$$s = L - 1 - r$$

Example

the following matrix represents the pixels values of an 8-bit image (r) , apply negative transform and find the resulting image pixel values.

Image (r)			
100	110	90	95
98	140	145	135
89	90	88	85
102	105	99	115

solution:

$$L = 2^8 = 256$$

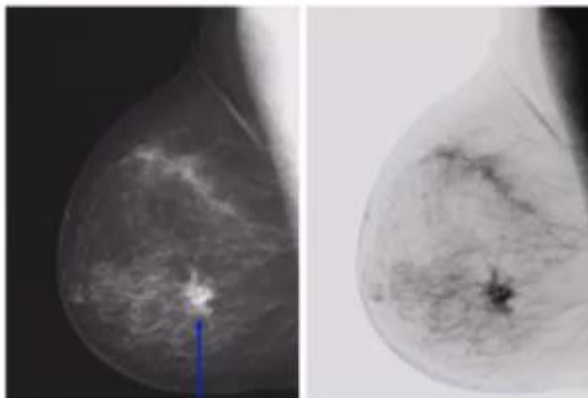
$$s = L - 1 - r$$

$$s = 255 - r$$

Apply this transform to each pixel to find the negative

Image (s)			
155	145	165	160
157	115	110	120
166	165	167	170
153	150	156	140

Image Negation (Example)



Shows a small lesion.

$$s = L - 1 - r$$

Produces an equivalent of a photographic negative.
Enhances white or gray detail embedded in dark regions.

FIGURE 3.4
(a) Original digital mammogram.
(b) Negative image obtained using the negative transformation in Eq. (3.2-1).
(Courtesy of G.E. Medical Systems.)

Log Transformation Function

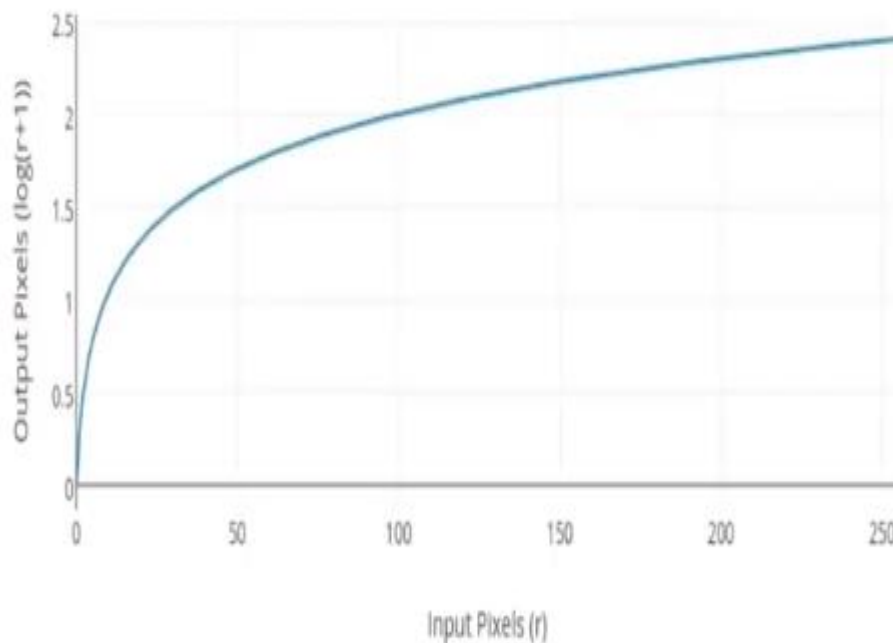
- The general form is:

$$s = c \log(1+r) \quad r \geq 0$$

- $(1+r)$: All the input intensity values have been incremented by 1 because the input values vary from 0 to 255 and the $\log(0)$ is not defined.
- r varies from (0-255), $\log(1+r)$ ranges from (0-2.4).
- The role of the multiplier C is to make the log-transformed pixel values span the entire range of 256 grayscale levels available for the output image.

$$255 = c \log(r_{\max} + 1) \quad c = 255 / \log(r_{\max} + 1)$$

Log Transformation Function



Log Transformation Function

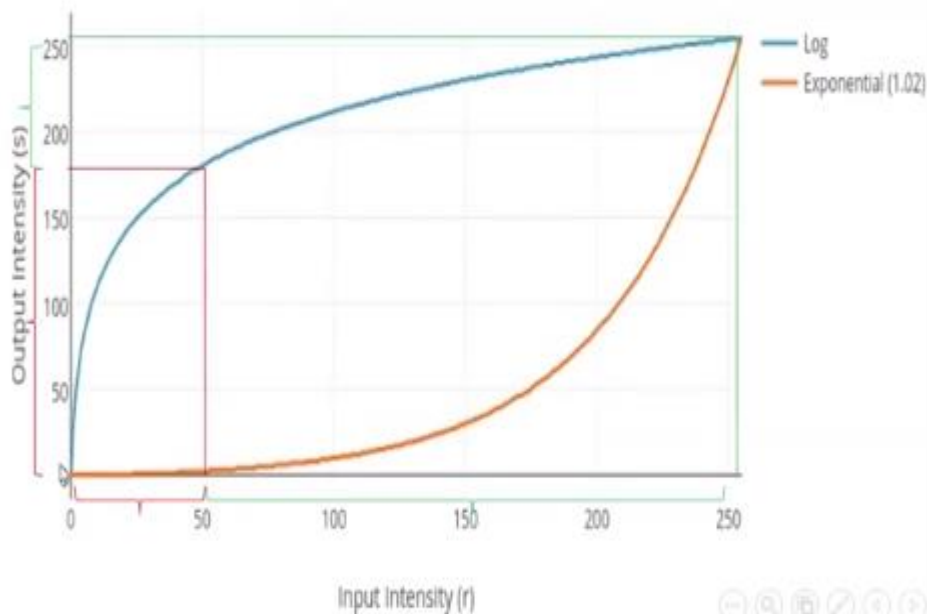
- The general form is:

$$S = c \log(1+r) \quad r \geq 0$$

where c is a constant, and it is assumed that

- It maps a narrow range of low gray-level values in the input image into a wider range of output levels.
- The opposite is true of higher values of input levels.
- Used to expand the values of dark pixels in an image while compressing the higher-level values.
- The opposite is true of the inverse log.
- The power-law transformations are much more versatile for this purpose than the log transformation

Log Transformation Function



Log Transformation Function

Input range



Output range after applying log transformation function



Intensity values of 0 and 15 in the input are mapped to 0 and 127 in the output.

The exact opposite phenomenon takes place at the other end of the spectrum. For example, pixels with intensities of 205 and 255 are mapped to 245 and 255 by the log transform.

Exponential or Inverse-Log Transformation

- The general form is:

$$s = T(r) = c(b^r - 1)$$

- Typically, b is chosen to lie close to 1.

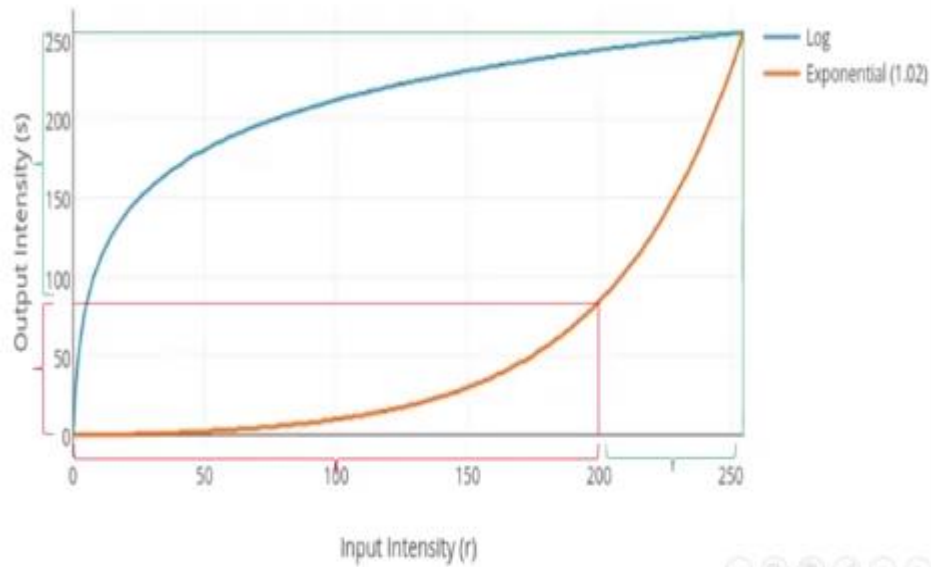
Input range



Output range after applying inverse log transformation function



Inverse Log Transformation Function



Example: Log Trans. Function



This example proves that the log transform can be effective in editing pictures that have been captured *against the light source* by digging out contrast information from the darker regions of an image at the cost of the brighter segments

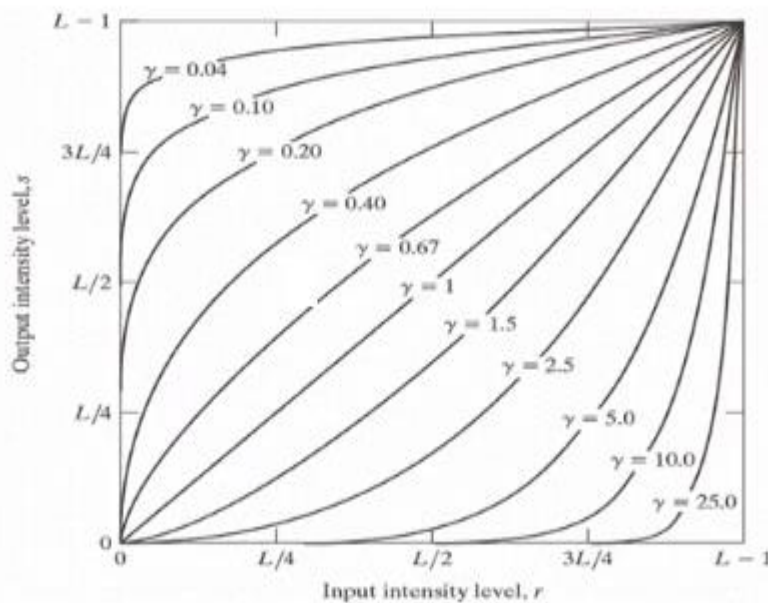
Power-Law Transformation

- Power-law transformations have the basic **form**

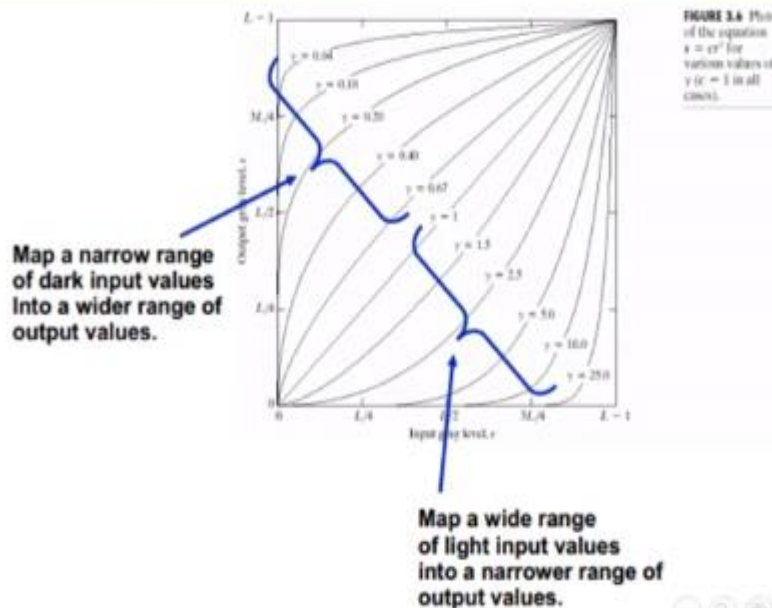
$$S = cr^\gamma$$

- Where c and γ are positive constants.
- Plots of s versus r for various values of γ are shown in **Fig. 3.6**.
- As in the case of the log transformation, power-law curves with fractional values of γ map a **narrow range** of dark input values into a **wider range** of output values, with the opposite being true for higher values of input levels.

FIGURE 3.6 Plots of the equation $s = cr^\gamma$ for various values of γ ($c = 1$ in all cases). All curves were scaled to fit in the range shown.



Power-law transformation



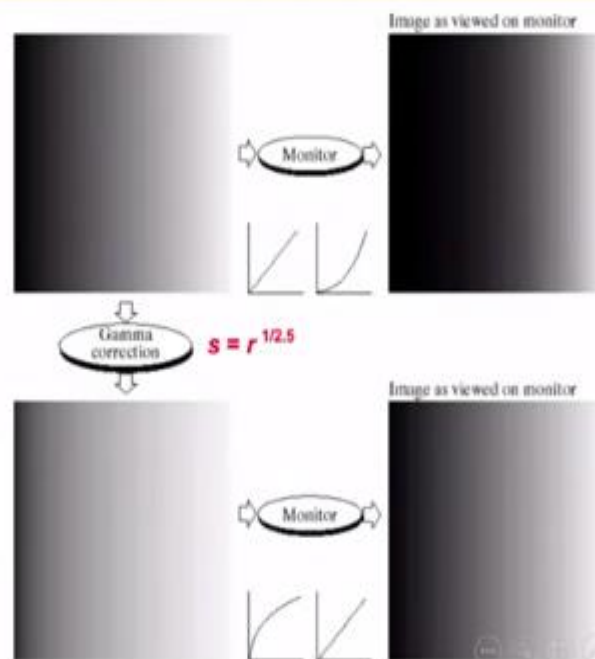
Power-Law Transformation

a b
c d

FIGURE 3.7
(a) Linear-wedge gray-scale image.
(b) Response of monitor to linear wedge.
(c) Gamma-corrected wedge.
(d) Output of monitor.

A variety of devices for image capture, printing and display respond according to power law.

Such systems tend to produce images that are darker than intended.



Example: Power-Law Transformation

Original image
is predominantly
dark!



FIGURE 3.8
(a) Magnetic resonance (MRI) image of a fractured human spine.
(b) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and $\gamma = 0.6, 0.4$, and 0.3 , respectively. (Original image for this example courtesy of Dr. David R. Pokuam, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

$$s = r^{0.6}$$

$$s = r^{0.4}$$

$$s = r^{0.3}$$